

Opinion Network Formation: Bipartite “Extremes” Sub-Report

Tanush Garg
IIIT Hyderabad

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Dataset Documentation: Interpretation and Preparation

The raw dataset comprises survey responses evaluating opinions across four primary domains: Technology, Education, Ethics, and Environment. The dataset contains 96 individual respondents and 60 distinct questions. The original responses were recorded using a standard 5-point Likert scale. To mathematically model the data as an ideological network, a multi-step data preparation pipeline was implemented.

Data Mapping and Ingestion

The text-based Likert responses were first mapped to a discrete numerical scale to facilitate algorithmic processing:

- Strongly Disagree $\rightarrow$ 1
- Disagree $\rightarrow$ 2
- Neutral $\rightarrow$ 3
- Agree $\rightarrow$ 4
- Strongly Agree $\rightarrow$ 5

Each respondent was assigned a unique `Student_ID`, and the dataset was transformed from a wide matrix into a long-format edge list consisting of `(Student_ID, Question_ID, Response)` tuples.

The “Extremes” Filter

To extract meaningful topological features rather than general consensus, moderate responses (2 and 4) and neutral responses (3) were systematically dropped from the edge list. By retaining only the absolute extremes (1 and 5), the dataset was refined to represent intense ideological commitment. This filtering process isolates the “gravity wells”, the specific issues that provoke the strongest polarized reactions from the cohort.

Application to the Network Model

The prepared edge list serves as the foundational architecture for an undirected bipartite graph. The network is constructed using two disjoint sets of nodes:

- **Set U (Respondents):** The 96 surveyed students.
- **Set V (Questions):** The 60 survey items.

An edge $E_{u,v}$ is drawn between a student $u \in U$ and a question $v \in V$ if and only if the student registered an extreme response to that specific question. This strict bipartite structure allows us to measure the direct degree of polarization for each question and enables the subsequent projection of a question-to-question network to map hidden ideological packages.

Pipeline Followed: Network Construction and Properties

To mathematically model the ideological intensity of the cohort, the filtered dataset was mapped into an undirected bipartite graph. The pipeline utilized the `NetworkX` library in Python to instantiate and validate the network architecture.

Mathematical Definition of the Graph

The network is defined as a bipartite graph $G = (U, V, E)$, where the vertices are divided into two disjoint and independent sets:

- **Set U (Students):** Representing the individual respondents.
- **Set V (Questions):** Representing the 60 distinct survey items.

The edge set E is defined such that an edge $e_{u,v}$ exists if and only if $u \in U$, $v \in V$, and the respondent u registered an absolute extreme response (Strongly Disagree or Strongly Agree) to question v . No edges exist between two students ($U \times U = \emptyset$) or two questions ($V \times V = \emptyset$) in this foundational projection.

Construction Pipeline

The graph construction followed a rigid three-step algorithmic pipeline:

1. **Data Ingestion & Standardization:** The raw tabular dataset was melted into a long-format edge list. Text-based Likert responses were mapped to a discrete numerical scale (1 to 5), standardizing potential capitalization inconsistencies.
2. **Threshold Filtering:** To eliminate low-conviction noise, all rows containing moderate (2,4) or neutral (3) responses were programmatically dropped. The remaining dataset consisted strictly of 1s and 5s.
3. **Bipartite Instantiation:** Nodes were added to the graph with explicit bipartite attributes (0 for students, 1 for questions). The filtered edge list of (u, v) tuples was then appended to the graph.

Observed Topological Properties

Upon successful execution of the construction pipeline, diagnostic functions were run to verify the structural integrity and baseline topological properties of the graph:

- **Node Distribution:** The network successfully compiled with **149 total active nodes**. This comprises all **60 question nodes** and **89 student nodes**. Notably, 7 of the original 96 students were mathematically filtered out of the network entirely, indicating they did not register a single extreme opinion across the survey.
- **Edge Density:** The graph contains **2,362 extreme opinion edges**, revealing a dense web of high-conviction stances.
- **Macro-Connectivity:** A critical structural diagnostic revealed that the network forms **exactly 1 connected component**. This mathematical property proves that despite the

filter restricting the graph exclusively to highly polarized opinions, there are no entirely segregated ideological echo chambers. Every radicalized stance is tethered to the rest of the cohort through shared “hub” questions, weaving the entire class into a single, cohesive macro-structure.

Analysis and Visualizations

Degree Distribution and Testing for Randomness

To determine the topological nature of the extreme ideological formations, the degree k of each question node was extracted. The degree directly quantifies the “gravity” of a question, specifically the number of students who registered a highly polarized response to it. Based on the survey data, the extreme response counts per question ranged from a minimum of 11 to a maximum of 64.

To classify the network structure, the degree distribution $P(k)$ was analyzed. Because the degree range is relatively narrow and does not span multiple orders of magnitude, standard **linear binning** was utilized to visualize the probability density without requiring logarithmic compression.

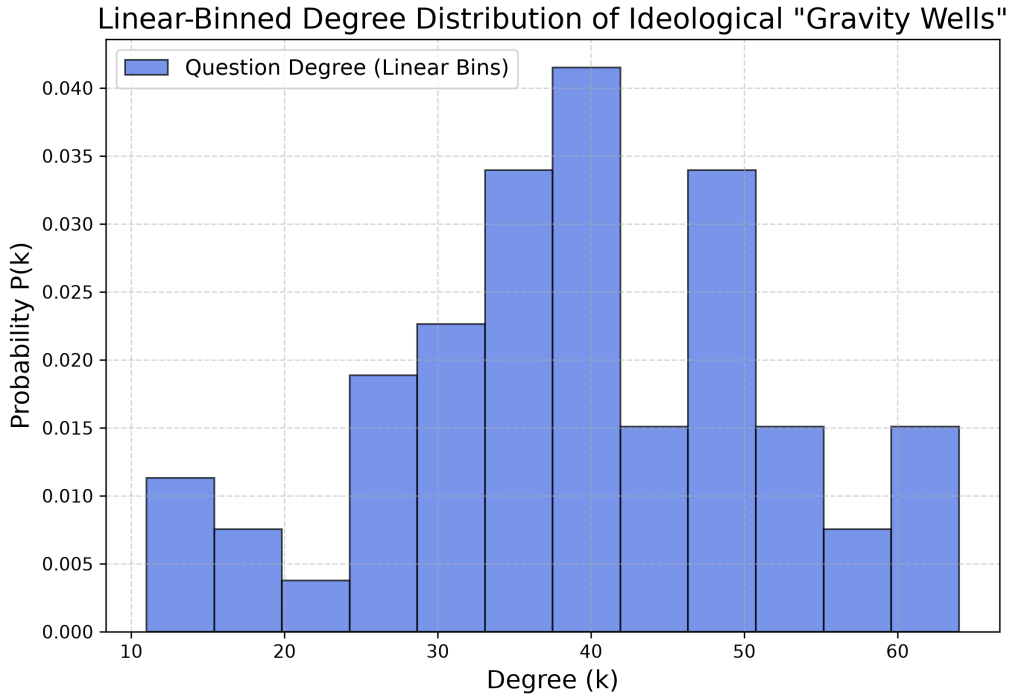


Figure 1: Linear-binned degree distribution of the question nodes.

Structural Interpretation

The resulting histogram (Figure 1) reveals the underlying mathematical nature of the class’s ideological structure:

- **Absence of a Scale-Free Topology:** The data does not exhibit the heavy, elongated tail indicative of a power-law distribution ($P(k) \propto k^{-\gamma}$). This indicates the network is *not* scale-free.
- **Erdős-Rényi Characteristics:** Instead, the distribution is visibly peaked in the center (around $k = 40$) and bounded on both ends, closely mirroring the Poisson limit of an

Erdős-Rényi (ER) random graph. In an ER model, the probability of a node possessing an extremely high degree is suppressed by a super-exponential factorial tail ($P(k) \sim \lambda^k/k!$), meaning that massive, disproportional “hubs” are effectively forbidden.

Consequently, the mathematical evidence suggests that while the class holds strong extreme opinions, these opinions are somewhat normally distributed across the 60 questions. No single issue monopolizes the entire cohort’s radical attention; rather, the “gravity wells” are distributed in a localized, predictable manner.

Centrality Metrics

To evaluate the topological influence of specific questions, three centrality measures are computed on the projected network:

- **Degree Centrality (C_D):** Identifies mainstream focal points that provoked the highest absolute volume of extreme reactions.
- **Betweenness Centrality (C_B):** Isolates questions acting as ideological bridges between otherwise disconnected student factions.
- **Eigenvector Centrality (C_E):** Identifies the core tenets of the most radicalized factions, calculating influence based on connections to other highly polarizing questions.

The Molloy-Reed Criterion

To mathematically determine whether the extreme ideologies form a connected giant component or remain as isolated, fragmented factions, the excess degree is calculated. The connectivity threshold is evaluated using the Molloy-Reed criterion:

$$\frac{\langle k^2 \rangle}{\langle k \rangle} > 2$$

Results and Discussion

[Placeholder: Insert interpretations here once the Jupyter Notebook executes the adjacency matrices. Highlight specific nodes (e.g., T14 on social media polarization or V05 on single-use plastics) that emerged as primary anchors. Discuss the exact numerical outcome of the Molloy-Reed criterion and what it reveals about the overarching ideological unity or fragmentation of the class.]